

Catherine Varas

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Profile

AI & Data Engineer focused on designing and implementing AI systems. I work across the full picture, from the data infrastructure that feeds the models to the orchestration layers that make them work in production. My experience spans computer vision pipelines for scientific research, RAG and automation systems for enterprise clients, and large-scale data architectures across different industries.

Having worked across data engineering, software architecture, and applied AI helps me design systems that are robust and actually hold up in production. I bring a research-driven approach to every problem, I learn the domain, understand the data, and deliver solutions that work reliably at scale. I'm looking for a role in AI, machine learning, or data architecture, in a team tackling meaningful problems.

Industry Experience

Data Engineer / Business Intelligence Analyst

Sep 2026 – Present

Harmony Australasia | Australia

- Designed and developed Azure Data Factory pipelines integrating operational databases, equipment telemetry, production systems and other enterprise sources into a centralised data lake for Power BI analytics and machine learning
- Implemented a Medallion Architecture (Bronze, Silver, Gold) to ingest, standardise, validate and transform raw operational data into trusted analytical datasets for reporting, advanced analytics and ML

Artificial Intelligence Engineer

Mar 2026 – Present

NINCA Tech Services | Freelance, Hybrid

- Designed and built an agentic automation platform from the ground up, integrating n8n and LLMs to orchestrate business processes and synchronise ERP systems, dashboards and real-time WhatsApp communication for operational teams
- Developed a custom operations management platform combining ERP functionality, workflow automation, centralised data and interactive dashboards into a single system
- Built an AI-powered analytics platform with a conversational interface that answers data questions and reconfigures visualisations on request, using RAG and LLMs with full query traceability

Software & Data Engineer

Mar 2026 – Sep 2026

Australian Institute for Bioengineering and Nanotechnology (AIBN) | Brisbane, Australia

- Developed an end-to-end data analysis and simulation system for bio-experiments, integrating bioreactor, sample and metabolomics data into validated metrics, visual insights and actionable results across multiple bioreactor platforms
- Built an interactive environment to analyse and simulate fermentation processes: explore experiments, modify parameters, recalculate results dynamically, compare runs and test scenarios in one workflow
- Engineered the platform with an AI-assisted development workflow (Claude Code), modular architecture, automated testing, data validation and reproducible pipelines; presented at GBA 2026 in France

Data & AI Engineer

Jul 2025 – Feb 2026

One51 | Data & Analytics Consultancy | Sydney, Australia

- Architected cloud-based AI systems on Azure (Machine Learning, Durable Functions, AI Search, Key Vault, DevOps CI/CD): scalable AI orchestration, HuggingFace model serving, Logic App integrations and cost-optimised workflows
- Developed an AI-powered migration platform combining metadata-driven processing and RAG to automate analytics migrations (QlikView to Power BI, SQL Server to Snowflake), reducing manual mapping and improving accuracy
- Engineered an AI-driven accounting automation system on Databricks combining OCR, signal normalisation and LLM-based semantic resolution, deployed through automated Databricks Jobs and pipelines
- Built analytics and migration-intelligence apps in Streamlit and Python: dependency mapping, migration-gap detection, automated scoring, validation and data-quality assessment

Associate Software & Data Architect

Jul 2024 – Jan 2025

BBVA | Lima, Peru

- Led migration of millions of contractual records from IBM Mainframe to cloud infrastructure using Apache Spark
- Designed frameworks for data quality, scalability, and compliance with banking regulatory standards
- Developed Java backend routines dynamically querying Mainframe or Oracle systems depending on data recency
- Managed centralized contracts module exposing data across banking systems

Software Architect

Oct 2022 – Jul 2024

BBVA | Lima, Peru

- Engineered scalable batch pipelines using Apache Spark and Java for multi-product Mainframe-to-cloud migration
- Developed reusable Spark routines for data transformation, validation, reconciliation, and loading in Parquet format
- Automated business-rule validations and anomaly detection across millions of contract records
- Deployed and maintained 70+ reusable components (Java backend, Spark batch, database scripts)

BI & Data Analyst

Jul 2021 – Feb 2022

Vida Software SAC | Lima, Peru

- Built Big Data pipelines for performance reporting; automated daily processing of 200+ SFTP files
- Designed dashboards for inventory and sales forecasting for multinational retail clients
- Built integration pipelines between Hasura GraphQL and SQL Server; updated Node.js/Vue.js components

Education

Master in Data Science

2025 – June 2026

University of Queensland | Brisbane, Australia

- Coursework: Cloud Computing, Machine Learning, Statistics, Artificial Intelligence, Deep Learning, Bioinformatics

Master of Artificial Intelligence

2024 - 2026

University of La Rioja | Spain

- Coursework: Generative AI, Machine Learning, Computer Vision, Neural Networks

Bachelor of Industrial Engineering

2019 – 2023

Peruvian University of Applied Sciences | Peru

- Coursework: Thermodynamics, Chemistry, Materials and Manufacturing Technology

Associate’s Degree in Computer Science

2019 – 2022

Institute Cibertec | Peru

- Coursework: SQL, Java, Python, C#, Oracle

Research Experience

Research Volunteer — Computer Vision

Nov 2025 – Present

UQ Queensland Brain Institute | Brisbane, Australia

- Built a multi-model pipeline (YOLO + SAM2) for automated animal tracking
- Engineered chunked GPU streaming and HPC parallel execution to process long videos without memory overflow
- Designed a geometric classifier detecting 6 social interaction types from spatial coordinates
- Reconstructed neural network architectures from raw checkpoint weights and debugged precision corruption

Technical Skills

Languages	Python, R, Java, SQL, Bash, C#, JavaScript
ML / AI	YOLO, SAM2, Neural Networks, HuggingFace, RAG, scikit-learn, Deep Learning, Computer Vision, Machine Learning
Data Engineering	Apache Spark, Databricks, ETL Pipelines, Parquet, GraphQL, Control-M
Visualization	Dash/Plotly, Power BI, Streamlit, matplotlib, Publication-ready Figures
Cloud & Infra	Azure (ML, Functions, AI Search, DevOps), AWS, GCP, HPC, GPU Streaming
Databases	Oracle, SQL Server, PostgreSQL, MySQL, Elasticsearch
Practices	Reproducible Research, Git, CI/CD (Jenkins, Azure DevOps), Modular Architecture, Testing

Certifications

- Generative AI Fundamentals — Academy Accreditation
- Microsoft Certified: Fabric Data Engineer Associate (DP700)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP600)
- Microsoft Certified: Power BI Data Analyst Associate (PL300)
- AWS Cloud Practitioner
- Microsoft Azure Fundamentals